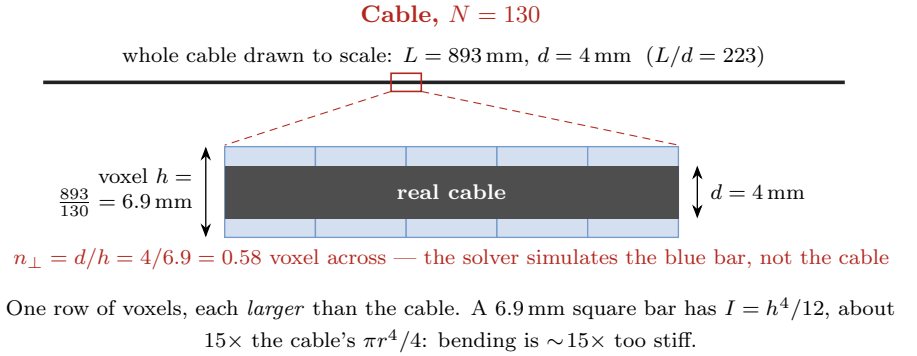
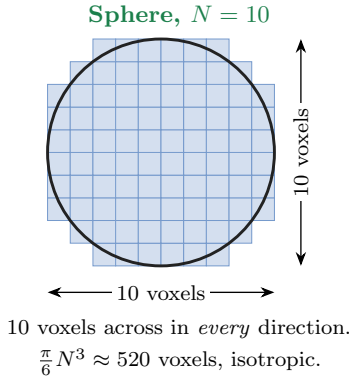


Why voxel FEM suits a sphere but not a cable

PhysX volumetric deformables in Isaac Sim — one-page note

Claim. Isaac Sim builds the FEM simulation mesh from *cubic voxels* of one size, $h = L_{\max}/N$, where $L_{\max}$ is the body's longest dimension and N the hexahedral resolution. The number of voxels across the *thinnest* dimension d is therefore $n_{\perp} = Nd/L_{\max}$. A sphere has $d/L_{\max} = 1$; our cable has $d/L_{\max} = 1/223$, so its cross-section — the only thing that sets bending stiffness — is left unresolved.



The two ways out, and why neither works

(cross-sections below share one scale)

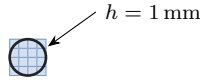
(a) As built: $N = 130$

(b) “Just add voxels”: $N = 893$

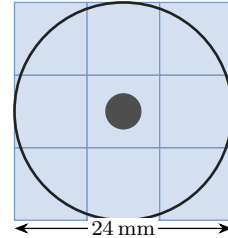
(c) Fatten the rod $6\times$ (our script)



1 voxel, $h = L/N = 6.9$ mm.
 Wrong cross-section:
 $EI \sim 15\times$ too high.



4 voxels across, but $893 \times 4^2 \approx 14000$ voxels and a chain 893 elements long. Iterative solver cannot converge along it: cable turns to rubber, or the step must shrink.



3 voxels across at $N = 112$; rescale $E \cdot q^4$, $\rho \cdot q^2$ ($q = \frac{1}{6}$) to keep EI and mass. EA becomes $36\times$ too soft and the contact shape is $6\times$ too big.

Calculations

Inputs: $L = 893$ mm (from the photo), $r = 2$ mm, $d = 4$ mm, $E = 80$ MPa, $\rho = 2390$ kg/m³ (`cable_config.py`); $N = 130$, $r_{\text{sim}} = 12$ mm, $\Delta t = 1/240$ s (`hang_physx_fem.py`).

Thin cable, $d = 4$ mm

aspect ratio	$L/d = 893/4 = 223$
voxel size	$h = L/N = 893/130 = 6.9$ mm
voxels across	$n_{\perp} = d/h = 4/6.9 = 0.58$
bending error	$\frac{h^4/12}{\pi r^4/4} = \frac{1.9 \times 10^{-10}}{1.3 \times 10^{-11}} \approx 15$
4 voxels across	$N = 4L/d = 4 \cdot 893/4 = 893$
number of voxels	$893 \cdot 4^2 \approx 14000$
wave speed	$c = \sqrt{E/\rho} = \sqrt{80 \times 10^6 / 2390} = 183$ m/s
crossing time	$h/c = 1 \text{ mm} / 183 = 5.5 \mu\text{s}$
step / crossing	$\Delta t / (h/c) = (1/240) / 5.5 \mu\text{s} \approx 760$

Fat rod, $r_{\text{sim}} = 12$ mm

radius ratio	$q = r/r_{\text{sim}} = 2/12 = 1/6$
resolution	$N = \lceil 3L/(2r_{\text{sim}}) \rceil = 112$, $h = 8.0$ mm
voxels across	$n_{\perp} = 24/8.0 = 3$
rescaled material	$Eq^4 = 62$ kPa, $\rho q^2 = 66$ kg/m ³
stretch stiffness	$EA: 1005 \text{ N} \rightarrow 28 \text{ N}$ ($q^2 = 1/36$)
Thick cable, $d = 20\text{--}30$ mm	
aspect ratio	$893/20 = 45$, $893/30 = 30$
voxels across	$20/6.9 = 2.9$, $30/6.9 = 4.4$
4 voxels across	$N = 4 \cdot 893/20 = 179$, $4 \cdot 893/30 = 120$
step / crossing	$6.9 \text{ mm} / 183 = 38 \mu\text{s}$, ratio ≈ 110

	Sphere	Thin cable ($d = 4$ mm)	Thick cable ($d = 20\text{--}30$ mm)
Longest / thinnest dimension	1	223	45–30
Voxels across the thin direction at $N = 130$	130	0.58	2.9–4.4
N needed for 4 voxels across	4	893	179–120
Element chain the solver must converge along	N	893	120–179
Result	good	bad	works

Conclusion. Voxel FEM fits compact bodies and thick cables, where the grid resolves the cross-section. A thin cable is a one-dimensional object and should be modelled as one — a Cosserat rod or a capsule chain — where the cross-section enters through EI , EA and GJ . The rod and capsule-chain models in this project reproduce the photographed cable to 33–75 mm RMS (`metrics.csv`).